

Input Parameters

Slope Angle	40°
Rainfall Intensity	10 mm/h
Soil Type	Rocky
Vegetation Cover	20%
Fault Zone Proximity	10 km
Land Use Type	Urban
Drainage Condition	Moderate

Risk Score

40.3

Moderate

RISK CATEGORY

The site presents Moderate landslide susceptibility (score: 40.3/100). 3 factor(s) are contributing to elevated risk and 2 factor(s) are providing stabilising influence.

Hazard Analysis

! Slope Angle	Moderate slope angle contributes a baseline level of gravitational stress.
% ¹ / ₄ Rainfall Intensity	Low rainfall intensity limits pore-water pressure build-up.
% ¹ / ₄ Soil Type	Soil type provides relatively high cohesion and resistance to shear failure.
! Drainage Condition	Moderate drainage partially mitigates rainfall-driven saturation.
% ² Fault Zone Proximity	Close proximity to a fault zone pre-conditions slopes through seismic shaking and rock fracturing.
% ² Vegetation Cover	Low vegetation cover provides minimal root reinforcement and allows high surface runoff.
% ² Land Use Type	Urban land use adds significant surface loading and reduces natural slope protection.

Mitigation Recommendations

- 1 Increase vegetation cover